

System Overview- Database of Pharmacy and Medication Inventory Control

System Access: The software is strictly local and only accessible via authorized pharmacy computers.

Geographical Scope: While initially focused on the Community of Madrid, the system architecture supports a national hierarchy: All cities in Spain -> All municipalities/towns within each city -> All pharmacies within each municipality.

Each pharmacy contains:

- A comprehensive list of medications categorized by target illnesses.
- A registry of clients linked to their purchase history (medications bought)

Proposed Database Entities

Client: id, name, nationality, age, municipalityId (FK).

Medication: id, name, targetIllness, ss, prescription.

Pharmacy: id, registrationNumber, phone, address, municipalityId (FK), photo.

Municipality: id, name.

Supplier: id, name, phone.

Inventory: id, pharmacyId (FK), medicationId (FK), quantityStock, expirationDate, price, supplierId (FK), minimumStock.

Purchase: id, clientId (FK), pharmacyId (FK), date, medicationId (FK), quantity, price.

Order: id, pharmacyId (FK), supplierId (FK), date, status, medicationId (FK), quantity.

REQUIREMENTS

Functional requirements:

1. The system must maintain a comprehensive list of all medications available across all pharmacies in the Community of Madrid.
2. The system must allow users to filter the medication list based on specific search criteria.
3. The system must validate entity attribute lengths before attempting a database insertion.
4. The system must enforce spatial boundaries by ensuring any new Pharmacy entity is strictly linked via a Foreign Key to a valid registered Municipality within the province of Madrid.
5. The system must allow the pharmacist to manually add stock quantities for new or existing medications.
6. The system must automatically deduct stock quantities when a medication is sold.
7. The system must generate an automatic notification when a medication is out of stock.

8. The system must allow users to access detailed information about any specific medication.
9. The system must be able to verify the real-time stock availability of a specific medication at a designated pharmacy.
10. The system must allow pharmacists to generate a purchase order to a supplier when stock levels fall below a predefined threshold.
11. The system must allow pharmacists to update medication information (price, expiration date, supplier, etc.).
12. The system must prevent stock quantities from becoming negative.
13. The system must track the expiration date of medications.
14. The system must notify pharmacists when medications are close to their expiration date.
15. The system must maintain a history log of inventory transactions (stock additions, sales, and adjustments).

USE CASES

Use Case 1: System Login

Primary Actor: Pharmacist. **Preconditions:** The actor must be using an authorized pharmacy computer and possess valid credentials (pharmacy registration number and password).

Postconditions: The pharmacist is authenticated and gains access to the medication inventory list.

Main Scenario:

- 1-The pharmacist enters the pharmacy registration number into the system.
- 2-The system prompts the user for their password.
- 3-The system validates the credentials. Upon successful validation, it displays the main medication list.

Extensions: Invalid Credentials: If the registration number or password is incorrect, the system displays an error notification and denies access.

Use Case 2: Search for a Specific Medication

Primary Actor: Pharmacist. **Preconditions:** The pharmacist is authenticated and inside the "Search Medication" menu. **Postconditions:** The system displays the medication details along with its real-time stock availability, associated pharmacies, their municipalities, and suppliers.

Main Scenario:

1. The pharmacist searches for the medication name.

2. The system verifies that the medication exists.
3. If it exists, the system displays all information regarding that medication.

Extensions: Medication Not Found: The system displays an error via notification if the medication does not exist.

Use Case 3: Increase Stock of an Existing Medication

Primary Actor: Pharmacist **Preconditions:** The pharmacist is previously authenticated; the medication exists and is registered in the inventory. **Postconditions:** The medication quantity in the inventory is successfully increased/updated.

Main Scenario:

1. The pharmacist accesses the inventory system.
2. Searches for the medication in the system.
3. Selects the "Increase Stock" option.
4. The system displays the current stock of the medication.
5. The pharmacist enters the number of units received.
6. The pharmacist confirms the operation.
7. The system validates the entered data.
8. The system updates the total quantity of the medication.
9. The system displays a confirmation message.

Extensions: Medication not found or invalid quantity (0 or less).

Use Case 4: Reduce Stock of a Sold Medication

Primary Actor: Pharmacist and System **Preconditions:** The medication exists in the inventory and there is sufficient stock available. **Postconditions:** The medication quantity is updated in the inventory; if the stock reaches 0, the system triggers a notification.

Main Scenario:

1. The pharmacist accesses the inventory system.
2. Searches for the medication in the system.
3. Select the "Reduce Stock" option.
4. The system displays the current stock of the medication.
5. The pharmacist enters the number of units sold.
6. The pharmacist confirms the operation.
7. The system validates the entered data.
8. The system updates the total quantity of the medication.

9. The system displays a confirmation or alert message.

Extensions: Medication not found or invalid quantity (0 or less).

Use Case 5: Add a Non-Registered Medication

Primary Actor: Pharmacist **Preconditions:** The pharmacist is previously registered and the medication does not yet exist in the inventory. **Postconditions:** The medication is correctly saved to the registry.

Main Scenario:

1. The pharmacist accesses the inventory system.
2. Selects the "Add New Medication" option.
3. Enters all medication characteristics.
4. The system displays the current stock (initial entry).
5. The system validates the entered data.
6. The system adds it to the medication registry.
7. The system displays a confirmation message.

Extensions: Expired medication or medication already exists.

Use Case 6: Delete an Existing Medication

Primary Actor: Pharmacist **Preconditions:** The pharmacist is previously registered and the medication exists in the inventory. **Postconditions:** The medication is successfully deleted.

Main Scenario:

1. The pharmacist accesses the inventory system.
2. Selects the "Delete Medication" option.
3. The pharmacist enters the data of the medication to be deleted.
4. The system validates the entered data.
5. The system removes it from the medication registry.
6. The system displays a confirmation message.

Extensions: Medication not found or error during deletion.

Use Case 7: Notify Inventory Shortage

Primary Actor: System. **Preconditions:** The medication is registered in the inventory, a minimum stock level (e.g., 5) is defined, and a stock update has occurred (sale, expiration date, etc.). **Postconditions:** The pharmacist receives a notification on the screen.

Main Scenario:

1. The stock of a medication is updated (e.g., after a sale).
2. The system calculates the current stock and compares it with the configured minimum stock.
3. If the current stock is 0 or $\leq$ minimum stock, the system generates an alert.
4. The system logs the alert.
5. The system sends the notification to the pharmacist.

Extensions: Minimum stock level not configured or error sending the notification.

Use Case 8: Order Medication from Supplier to Restock

Primary Actor: Pharmacist. **Preconditions:** The pharmacist is authenticated, the medication exists in the inventory, and the medication has an assigned supplier. **Postconditions:** The order is registered with a status of "Sent" or "Pending." An order number is generated.

Main Scenario:

1. The pharmacist accesses the inventory and opens the "Orders" section.
2. Selects the medication to restock.
3. The pharmacist enters the quantity to order and confirms the supplier.
4. The system displays an order summary (medication, quantity, estimated price, delivery address).
5. The pharmacist confirms "Send Order."
6. The system validates data (quantity > 0, supplier availability, etc.).
7. The system registers the order with: Order ID, medication(s) and quantities, supplier, date/time, and initial status (Sent/Pending).
8. The system displays a confirmation.

Extensions: Invalid quantity, medication does not exist, or supplier does not exist.

Use Case 9: Generate Movement History

Primary Actor: Pharmacist and System **Preconditions:** The pharmacist is authenticated and registered movements exist (otherwise, the system shows an empty history). **Postconditions:** The system displays the list of movements according to applied filters.

Main Scenario:

1. The actor accesses the "History" section.
2. The system displays the history with the most recent movements by date.
3. The actor selects a movement to view details.
4. The system displays full movement information: Movement ID, date/time, medication, pharmacy, and movement type.

Extensions:

- No records found: If there are no movements matching the search criteria, the system displays a notification indicating that no history is available.
- Invalid filter values: If the pharmacist enters invalid data, the system shows an error message and requests corrected input
- Database error: If the system cannot retrieve the data, it displays an error notification

Traceability matrix:

Use Case	REQ1	REQ2	REQ3	REQ4	REQ5	REQ6	REQ7	REQ8	REQ9	REQ10	REQ11	REQ12	REQ13	REQ14	REQ15
C1	✓			✓											
C2	✓	✓						✓	✓		✓		✓		
C3					✓							✓			✓
C4						✓	✓					✓			✓
C5			✓		✓						✓		✓		✓
C6															✓
C7							✓			✓					
C8			✓							✓					✓
C9			✓		✓	✓									✓

Id (PK)	Name
MUN-0	Boadilla del monte
MUN-1	Pozuelo de Alarcon
MUN-2	Alcorcón

Pharmacy

Id (PK)	address	phone	registrationNumber	Municipality (FK)
P-0	Monteamor, 3	639826101	28-82016	MUN-0
P-1	Tomillo, 9	619362434	28-46194	MUN-1
P-2	Sotavento, 22	610372947	28-98451	MUN-2

Supplier

id	name	phone
S-0	Bayer	623401393
S-1	Cardinal Health	629461032
S-2	Bidafarma	64928351

Medication

id	name	targetIllness	ss	prescription
M-0	ibuprofeno	anti-inflammatory	FALSE	FALSE
M-1	paracetamol	analgesic/antipyretic	FALSE	FALSE
M-2	aspirina	Antiplatelet	FALSE	FALSE

Purchase

id	clientId (FK)	pharmacyId (FK)	date	medicationId (FK)	quantity	price
PUR-0	C-02	P-0	2026/04/12	M-1	1	4
PUR-1	C-01	P-1	2026/01/04	M-2	2	6
PUR-2	C-00	P-2	2026/03/26	M-0	1	5

Client

Id (PK)	name	nationalId	age	MunicipalityId (FK)
C-00	Paola	27673831L	70	MUN-1
C-01	Alex	36219443S	32	MUN-2
C-02	Alba	58347523P	15	MUN-0

Inventory

id	PharmacyId (FK)	MedicationId (FK)	SupplierId (FK)	stockQuantity	price	expirationDate	minimumStock
I-0	P-2	M-2	S-0	30	6	04/2029	7
I-1	P-0	M-1	S-2	20	4	03/2027	7
I-2	P-1	M-0	S-1	25	5	09/2032	7

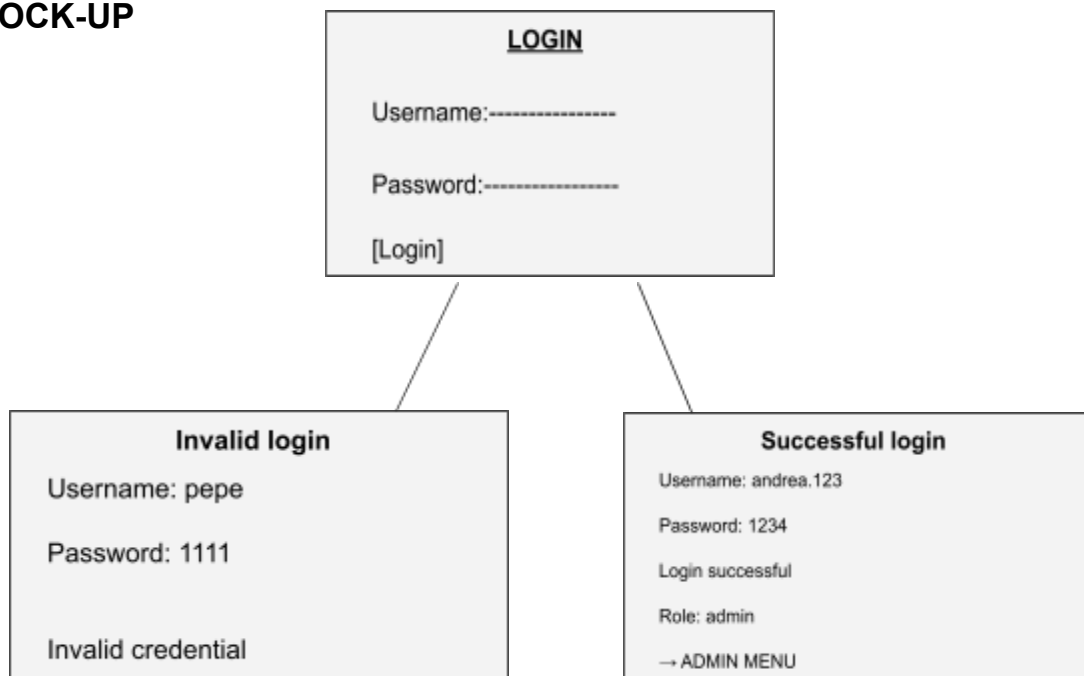
Order

id	PharmacyId (FK)	SupplierId (FK)	MedicationId	date	quantity	status
ORD-0	P-1	S-1	M-2	2026/03/12	10	pending
ORD-1	P-2	S-2	M-0	2026/01/10	15	pending
ORD-2	P-0	S-0	M-1	2026/02/11	20	pending

History

id	PharmacyId (FK)	MedicationId (FK)	movement Type	quantity	price	date
H-0	P-0	M-1	sale	1	4	2026/04/12
H-1	P-1	M-0	restock	25	125	2026/03/01
H-2	P-2	M-2	restock	10	30	2026/03/12

MOCK-UP



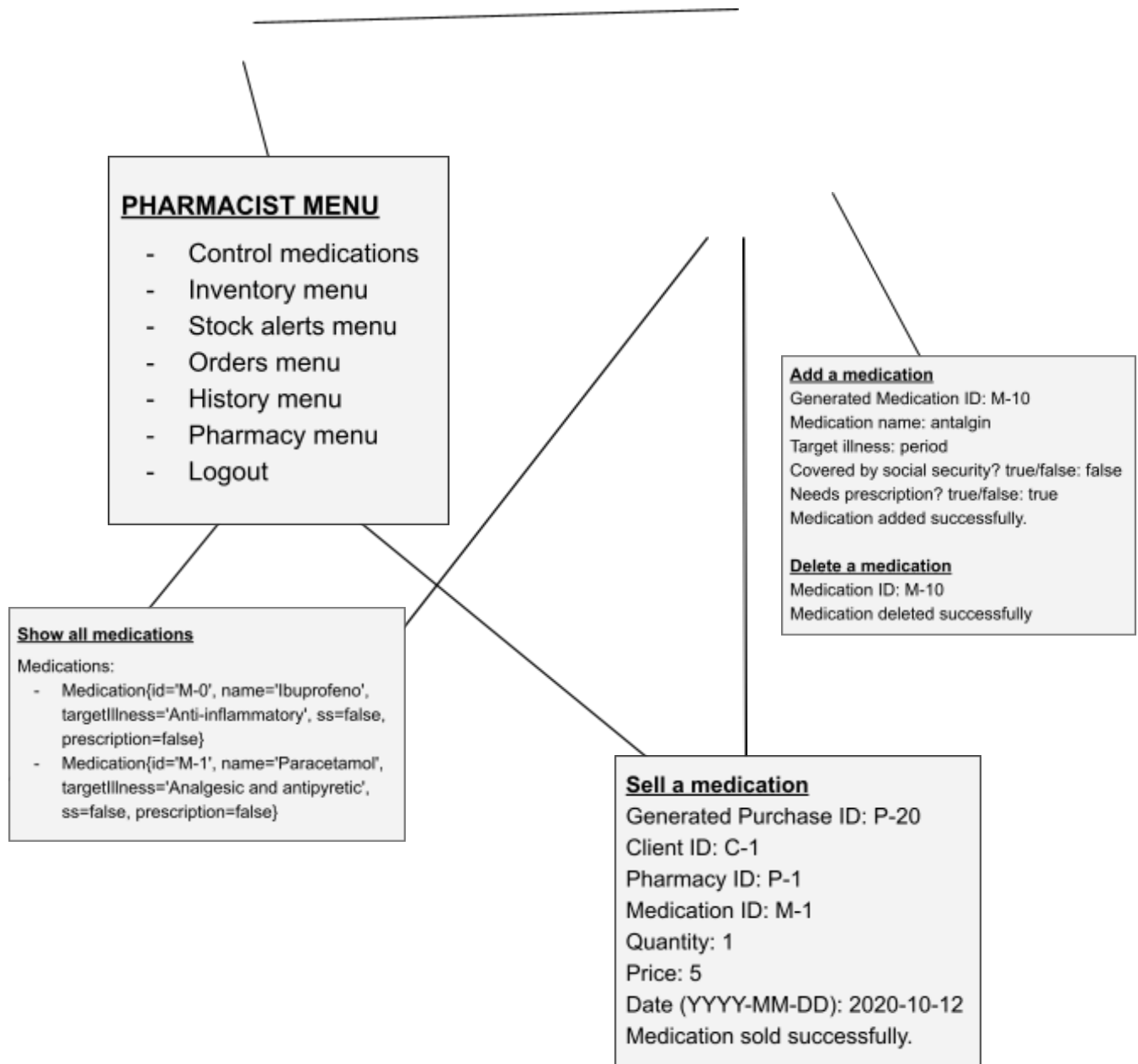
Search medication

Introduce medication name: [_____]

Medication{id='__', name='__',
targetIllness='__', ss=false/true,
prescription=false/true}

ADMIN MENU

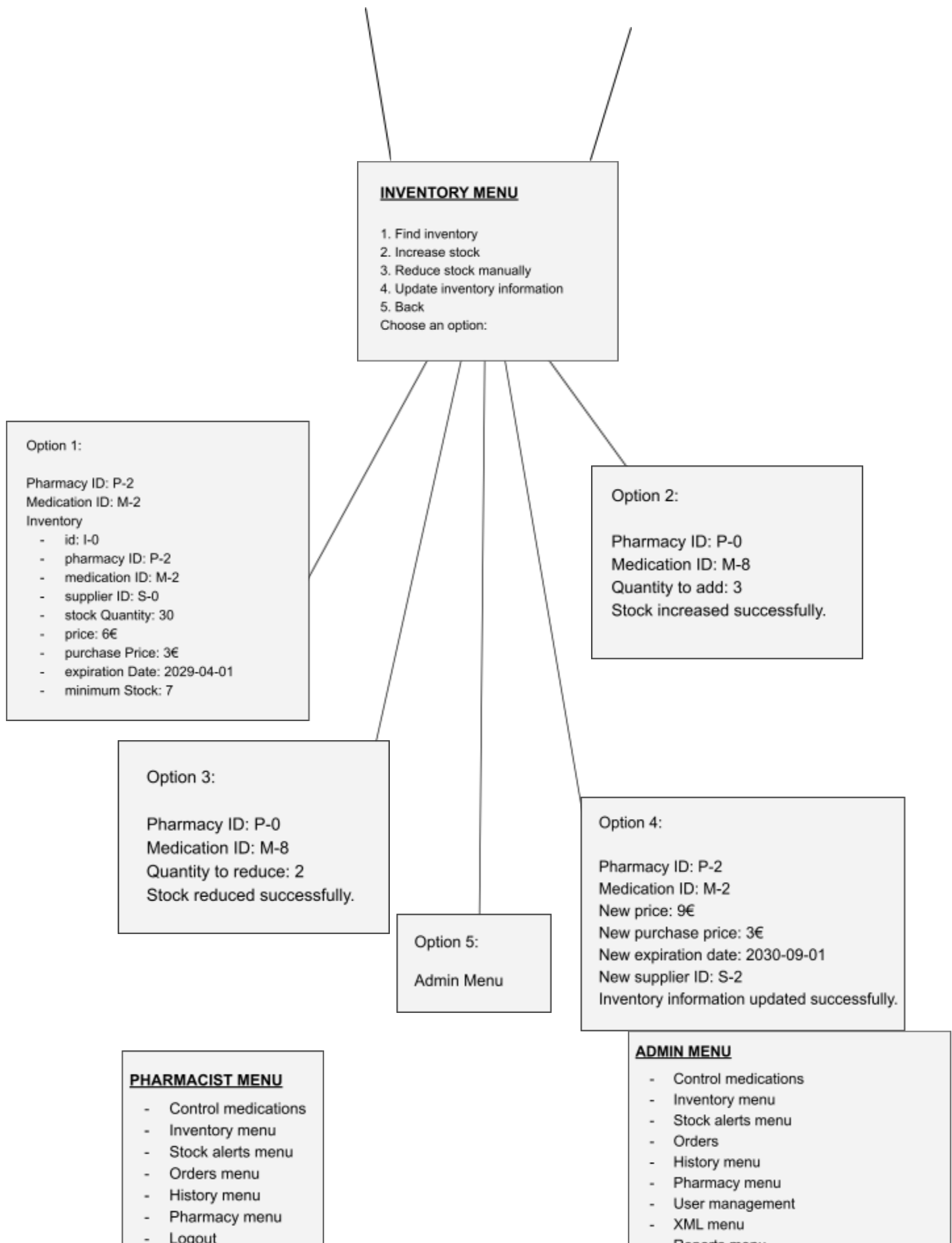
- Control medications
- Inventory menu
- Stock alerts menu
- Orders
- History menu

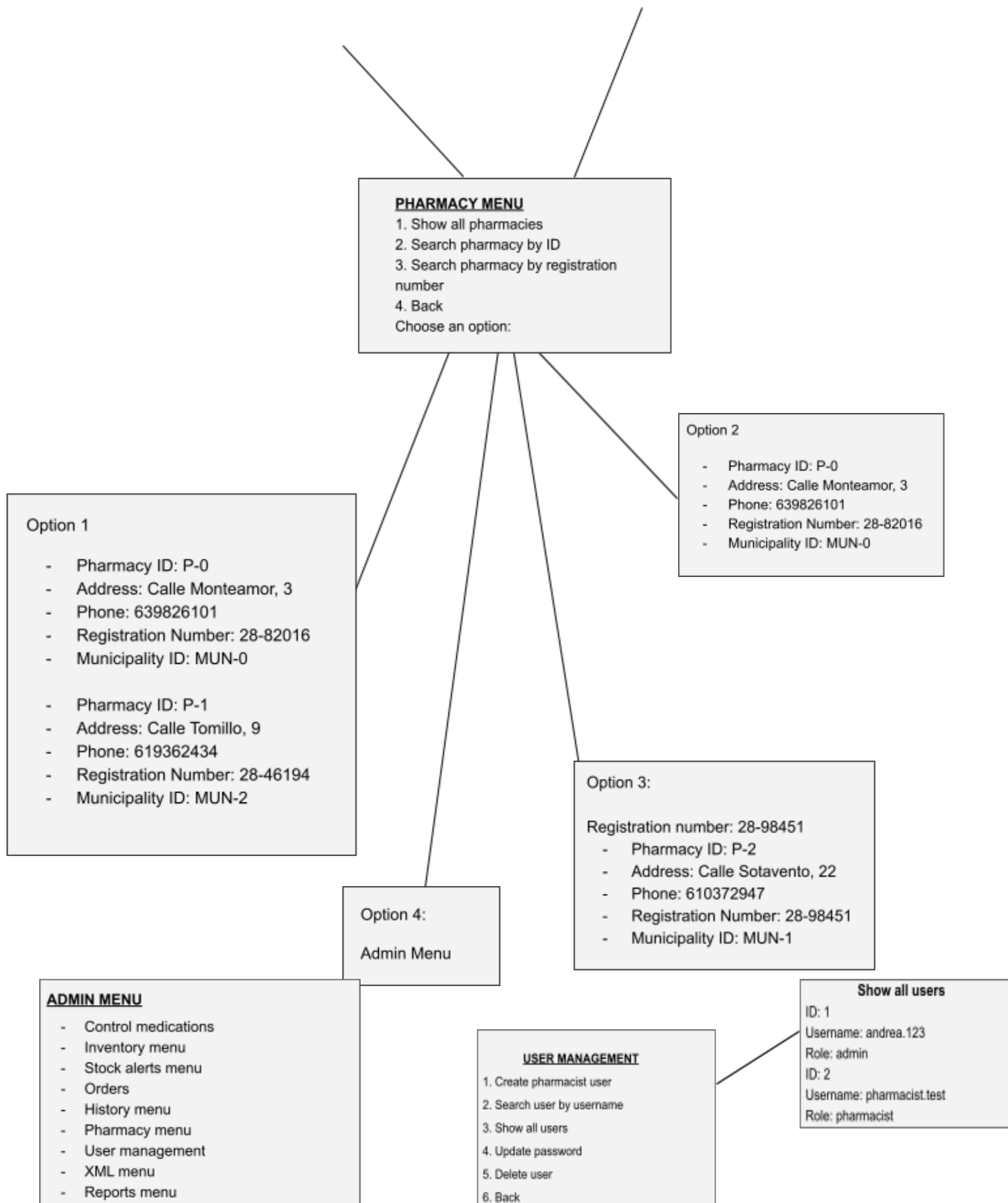
**PHARMACIST MENU**

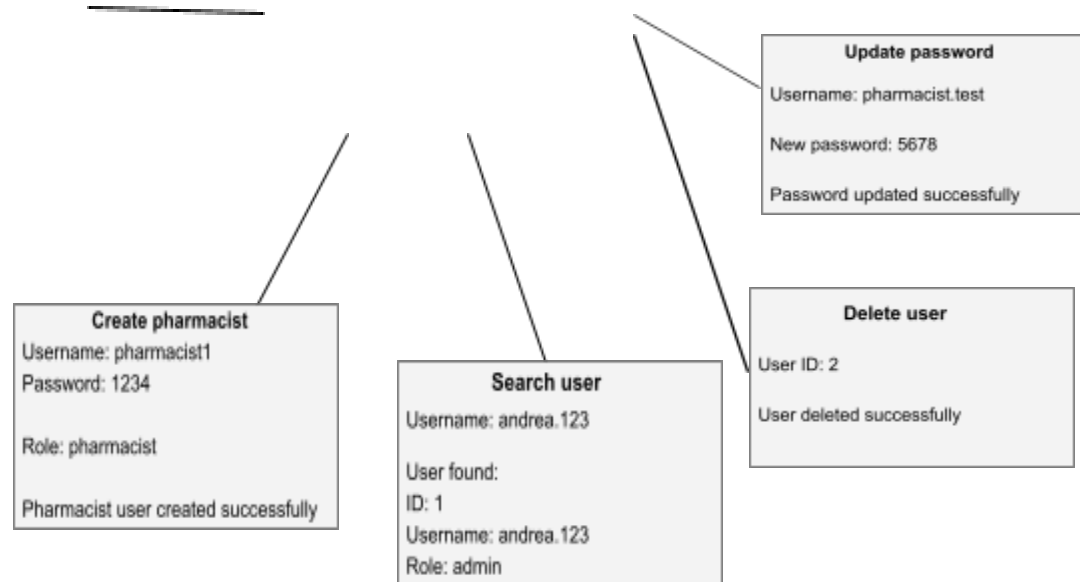
- Control medications
- Inventory menu
- Stock alerts menu
- Orders menu
- History menu
- Pharmacy menu
- Logout

ADMIN MENU

- Control medications
- Inventory menu
- Stock alerts menu
- Orders
- History menu
- Pharmacy menu
- User management
- XML menu
- Reports menu







ADMIN MENU

- Control medications
- Inventory menu
- Stock alerts menu
- Orders
- History menu
- Pharmacy menu

PHARMACIST MENU

- Control medications
- Inventory menu
- Stock alerts menu
- Orders menu

